Retrieval-Augmented Generation Assessment (RAGAS)

Evaluation set-up

The RAGAS framework (Es et al., 2024) was used to evaluate the quality of Bintanong's retrieved context and generated answers. The evaluation used the 14 document-based (RAG) benchmark queries, each asked in English, Tagalog and Taglish, for 42 query–answer pairs. These are the same 42 saved answers used for the Accuracy, Precision, Recall and F1-score evaluation, so no answer was regenerated for this measurement. RAGAS uses a separate large language model as a judge. DeepSeek (`deepseek-chat`) served as the judge, and the SEA-LION-E5-Embedding-600M model computed sentence embeddings. Each query has one reference answer, taken verbatim from the official Student Handbook or Citizen's Charter. It consists of every passage judged relevant to that query during the retrieval evaluation. The three language versions of a query share the same reference.

Five RAGAS metrics were computed. Context Precision and Context Recall evaluate the retrieval stage. Faithfulness, Answer Relevancy and Factual Correctness evaluate the generation stage.

Metrics and formulas

Faithfulness. This metric measures whether the generated answer stays within the evidence given to the generator, and therefore indicates hallucination. The judge breaks the answer into individual statements and decides for each one whether the evidence supports it.

Faithfulness = (number of answer statements supported by the evidence) / (total number of answer statements)

In Bintanong, a retrieved passage is never given to the generator alone. When the source documents were chunked and reviewed, each passage that cannot be understood without other passages recorded them as *required neighbours*. Examples are a procedure that continues across pages, or a penalty row that depends on a shared clause. When a passage is retrieved, its required neighbours are retrieved with it. The group reaches the generator only if it is complete, as reviewed fields with their labels and any recorded source limitation. Faithfulness was therefore measured against this full evidence, which was rebuilt from the deployed knowledge base. For all 42 answers, the rebuilt number of neighbours matched the number logged when the answer was generated.

Answer Relevancy. This metric measures whether the answer addresses the question asked. The judge writes N = 3 questions that the answer would respond to. The score is the average

cosine similarity between the embedding of each generated question (E_gi) and the embedding of the original question (E_o):

$$\textit{Answer Relevancy} = (1/N) \times \Sigma_{i=1..N} \cos(E_{gi}, E_o)$$

If the judge labels the answer as *noncommittal* (evasive or uncertain), the score is set to 0.

Context Precision. This metric measures whether the relevant passages are ranked above irrelevant ones in the top K retrieved results. For each rank k, v_k = 1 if the passage at rank k is relevant to the reference and 0 otherwise.

$$\textit{Precision@k} = \textit{relevant passages within the top k} / k$$

$$\textit{Context Precision@K} = \Sigma_{k=1..K} (\textit{Precision@k} \times v_k) / (\textit{number of relevant passages in the top K})$$

Context Recall. This metric measures whether retrieval found the information that the reference answer requires. The reference is split into statements, and each one is checked against the retrieved passages.

$$\textit{Context Recall} = (\textit{reference statements found in the retrieved passages}) / (\textit{total reference statements})$$

Factual Correctness (precision mode). This metric measures how much of the answer agrees with the reference answer. Answer claims supported by the reference are true positives (TP); claims not supported by it are false positives (FP).

$$\textit{Factual Correctness (precision)} = TP / (TP + FP)$$

Precision mode was used because the references are complete source passages. In recall or F1 mode, a correct but concise answer would be penalised for not repeating every sentence of the reference.

Worked example

Query 9 (English) is used to illustrate the computation: *"What is the penalty for cheating in an exam?"* Retrieval returned two passages, both from the Student Handbook: the penalty row for "a. Any form of cheating in examinations" (rank 1) and the clause "Additional corrective measures for all acts of academic dishonesty" (rank 2). Bintanong answered:

> Here is what the source says: Cause: "a. Any form of cheating in examinations"; 1st Offense: "Automatic grade of zero (0) on the examination/ activity or failing grade of five

> (5) upon discretion in the subject where it was committed."; 2nd Offense: "Suspension for two weeks"; 3rd Offense: "Suspension for one month"; Subsequent Offenses: "Dismissal". … Measures: "Withdrawal of degree; Withdrawal of honors; Disqualification from graduation with honors; Cancellation of registration". Recorded source limitation: "Unavailable information: which additional corrective measures apply automatically to an individual act." Please confirm with the relevant university office.

Faithfulness. The judge extracted 13 statements. Twelve were supported by the evidence, for example "The second offense results in suspension for two weeks", which is supported by the reviewed field "2nd Offense: Suspension for two weeks". The unsupported statement was the closing referral, "Please confirm with the relevant university office". Bintanong adds this sentence by design, and it does not appear in the source. Faithfulness = 12 / 13 = **0.923**.

Answer Relevancy. The judge generated three questions from the answer. Their cosine similarities to the original question were:

Generated question	cos(E_gi, E_o)
What are the penalties for cheating in examinations?	0.957
What are the penalties for cheating in examinations and additional corrective measures for academic dishonesty?	0.843
What are the penalties for cheating in examinations and the additional corrective measures for academic dishonesty?	0.838

Answer Relevancy = (0.957 + 0.843 + 0.838) / 3 = **0.879**. Question generation is not deterministic. The scored run, reported in Table R2, gave 0.840 for this item.

Context Precision. Both retrieved passages were relevant (v_1 = v_2 = 1), so Precision@1 = 1/1 and Precision@2 = 2/2. Context Precision = (1 × 1 + 1 × 1) / 2 = **1.000**.

Context Recall. The reference for Query 9 has four Handbook passages (pp. 88–90): the general cheating row and the additional corrective measures, which were retrieved, and two other examination-cheating rows (items e and g), which were not. The judge split the reference into 13 statements and found 9 in the retrieved passages. Context Recall = 9 / 13 = **0.692**. The missing statements are the penalties for items e and g, which shows that retrieval returned the general rule but not the related specific offences.

Results

Table R1. RAGAS scores of Bintanong by query language (n = 42; 14 per language)

Metric	Overall	English	Tagalog	Taglish
Faithfulness	0.883	0.891	0.898	0.858
Answer Relevancy	0.584	0.657	0.547	0.549
Context Precision	0.896	0.911	0.909	0.869
Context Recall	0.754	0.793	0.737	0.733
Factual Correctness (precision)	0.570	0.571	0.570	0.568

Table R2. RAGAS scores per benchmark query (mean of the English, Tagalog and Taglish versions)

No.	Query (English version)	Faithfulness	Answer Relevancy	Context Precision	Context Recall	Factual Correctness
1	How to drop subjects?	0.902	0.567	0.722	0.689	0.607
2	How to shift/change curricular program?	0.858	0.704	0.833	1.000	0.787
3	How to enroll for the next semester?	0.817	0.641	0.389	0.755	0.447
4	How to add or change subjects?	0.958	0.666	0.889	0.581	0.270
5	How to remove a grade of 4?	0.945	0.000	0.944	0.354	0.513
6	How to complete an INC grade?	0.930	0.000	1.000	1.000	0.430
7	How do I get off probation?	1.000	0.810	1.000	1.000	1.000
8	What do I do when I lose, forget my ID or to retrieve a confiscated ID?	0.800	0.835	1.000	1.000	0.890
9	What is the penalty for cheating in an exam?	0.915	0.767	1.000	0.692	0.387
10	How to secure a good moral certificate?	0.993	0.618	1.000	0.659	0.353
11	How to secure transfer credentials?	0.900	0.679	0.889	0.639	0.410
12	How do I claim accident insurance?	1.000	0.873	1.000	1.000	1.000
13	How to obtain a vehicle or gate pass?	0.646	0.374	1.000	0.380	0.430
14	How to avail the use of university rooms and facilities?	0.690	0.643	0.880	0.812	0.450
	Mean	**0.883**	**0.584**	**0.896**	**0.754**	**0.570**

Analysis of low-scoring items

- *Answer Relevancy of 0 (Queries 5 and 6, and Query 13 in Taglish).* These answers quote the source and then state a recorded source limitation (for example, conflicting completion deadlines for an INC grade) and refer the student to the relevant office. The judge labelled them noncommittal, which sets the score to 0.
- *Query 13 (vehicle or gate pass).* The source describes only selected office steps of the application, so the answer adds notices such as "These are selected office steps, not a

complete application procedure." These notices are not in the source text, which lowers Faithfulness. The gate-pass passages that were not retrieved also lowered Context Recall (0.380).

- *Query 3 (enrolment).* Context Precision was 0.389. The RAGAS judge considered some higher-ranked passages not useful for reaching the reference answer, although the human relevance judgments used for the F1 evaluation marked all three as relevant. RAGAS judges usefulness for the reference, not topical relevance, so the two measures can disagree.

Discussion

Faithfulness was high, 0.883 overall and at least 0.858 in every language. Answers stayed close to the institutional documents, which supports the design choice of quoting reviewed source fields instead of letting the model paraphrase freely. Most of the statements judged unsupported are notices Bintanong adds deliberately, such as source-limitation statements and referrals to the relevant university office. These follow the study's source-gap rules even though they are not in the source text.

Context Precision (0.896) shows that relevant passages were usually ranked first. Context Recall (0.754) shows that the retrieved passages usually covered most of the reference, but related clauses were sometimes missed, as in the worked example. Context Recall is likely overstated, because only passages that appeared in some system's top results were judged for relevance.

Answer Relevancy was the lowest metric (0.584). Seven answers scored 0 because the judge labelled them noncommittal: Queries 5 and 6 in all three languages, and Query 13 in Taglish. Each of these answers quotes the source and then states a recorded source limitation, such as conflicting completion deadlines, and refers the student to the relevant office. RAGAS treats such a hedge as evasive, while the study requires it whenever the source is incomplete or conflicting. Without these seven items, Answer Relevancy was 0.701 (English 0.766, Tagalog 0.638, Taglish 0.698). English scored higher than Tagalog and Taglish on this metric. Part of the gap may come from comparing a Filipino question with questions generated from mostly English source wording, so it is not a clean comparison of language quality.

Factual Correctness (0.570) was moderate. An answer claim counts against this score whenever it is absent from the reference, even if it is correct. Examples are content drawn from required-neighbour passages outside the reference, or system notices such as source-limitation

statements. The lowest items (Queries 4, 9 and 10) were not analysed claim by claim, so this metric is reported without attributing its errors to a specific cause.

Limitations: each item was scored once by a single judge model. The judge's reliability on Tagalog and Taglish was not separately validated. Query 11 was run on a later routing version; without it, the Table R1 scores change by at most 0.012. The planned comparison against the baseline Bintu-1 model was not carried out for RAGAS.

Chapter 5 entries

Summary of Findings (RAGAS)

- Bintanong's answers were highly faithful to the institutional documents: Faithfulness 0.883 overall (English 0.891, Tagalog 0.898, Taglish 0.858). Most unsupported statements were deliberate source-limitation notices and office referrals, not invented content.
- Retrieval ranked relevant passages first (Context Precision 0.896) and covered most of the needed information (Context Recall 0.754). Related clauses were sometimes missed.
- Answer Relevancy was 0.584 overall, or 0.701 without seven answers that RAGAS scored 0 for stating documented source limitations. English answers scored higher than Tagalog and Taglish answers.
- Factual Correctness against full source passages was 0.570 (precision mode).

Conclusion (RAGAS)

Bintanong's generation stage keeps its answers grounded in the reviewed PalSU documents in all three supported input languages, which meets the study's aim of a hallucination-resistant advisory system. The main measured weaknesses are incomplete retrieval of related clauses, and a mismatch between generic relevance metrics and the system's required disclosure of source gaps.

Contributions

- An evaluation procedure for neuro-symbolic RAG systems that scores faithfulness against the complete evidence the generator receives (retrieved passages plus required neighbours, as reviewed fields).
- Shared verbatim source references for 14 PalSU queries in three languages, usable by RAGAS and BERTScore alike.

Recommendations

- Add retrieval of related clauses within the same policy section (for example, other examination-cheating offences) to raise Context Recall.
- Report Answer Relevancy both with and without answers that disclose documented source limitations. Future work could adapt the noncommittal criterion so that a cited source gap is not treated as evasion.
- Repeat the RAGAS scoring with several judge runs or judge models, validate the judge on Filipino text, and include the baseline Bintu-1 comparison planned in Chapter IV.
- Extend the reference set to cover more queries per category, toward the 50-query benchmark planned in Chapter IV.